25901392 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEMATA00 - INTRODUCTION TO MACHINE LEARNING

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

One or two sentences

Answer any **5** questions

Each question carries **2** marks

1. Explain learning by self. [U] / [CO1]
2. What is scatter plot? [U] / [CO2]
3. What are predictive models? [U] / [CO3]
4. Write down the formula to evaluate F-measure. [K] / [CO3]
5. Give an example of a five dimensional data set. [U] / [CO4]
6. What is mean by "irrelevant variables" in the context of unsupervised machine learning task. [U] / [CO4]
7. Give the general form of a multiple linear regression model. [K] / [CO5]
8. What is a scatter graph? [K] / [CO5]

[2x5 = 10]

Part B

Short answer

Answer any **5** questions

Each question carries **6** marks

9. What are the possible ethical issues of machine learning applications? [K] / [CO1]

10. Explain quantitative data.What are the different types of quantitative data? [K] / [CO2]

11. Explain bias-variance trade-off in context of model fitting. [K] / [CO3]

12. Explain the external method approach for cluster quality evaluation. [K] / [CO3]

13. Illustrate the transformation of numeric(continuous) features to categorical features with an example. [A] / [CO4]

14. Explain the wrapper approach of feature selection. What are the merits and de-merits of this approach? [U] / [CO4]

15. Draw a scatter plot graph for the following data, indicating the relationship between internal marks and external marks of 15 university students: [U] / [CO5]

Internal Marks	15	23	18	23	24	22	22	19	19	16	24	11	24	16	23
External Marks	49	63	58	60	58	61	60	63	60	52	62	30	59	49	68

16. Explain subset selection approach to improve the accuracy of the linear regression. [U] / [CO5]

[6x5 = 30]

Part C

Essays

Answer any **3** questions

Each question carries **10** marks

17. Explain qualitative and quantitative data in details. Differentiate between the two. [K] / [CO1]

18. Write short notes on: [K] / [CO2]
    1) Nominal and ordinal data
    2) Box plot and histogram
    3) Mean and median

19. Evaluate the accuracy, the error rate, F measure and sensitivity of the model given the confusion matrix of the win/loss prediction of cricket match problem. [U] / [CO3]

	Actual Win	Actual Loss
Predicted Win	85	4
Predicted Loss	2	9

20. Write short notes on: [U] / [CO3]
    i. Bootstrap sampling
    ii. Lazy learner

21. [A] / [CO4]
    1. How can we calculate Hamming distance? Find the Hamming distance between 10001011 and 11001111.
    2. Compare the Jaccard index and similarity matching coefficient of two features having values (1, 1, 0, 0, 1, 0, 1, 1) and (1, 0, 0, 1, 1, 0, 0, 1).

22. Discuss maximum likelihood estimation in detail. [U] / [CO5]

[10x3 = 30]